



UNIVERSITY OF MUMBAI

Community Engagement Project Report On:-
Air and Noise Pollution Mapping

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

MR. RAKESH

ACADEMIC YEAR 2026-27



V. K. KRISHNA MENON COLLEGE

OF COMMERCE & ECONOMICS AND SHARAD SHANKAR DIGHE COLLEGE OF SCIENCE

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BHANDUP EDUCATIONAL SOCIETY'S
V. K. KRISHNA MENON COLLEGE

OF COMMERCE & ECONOMICS AND SHARAD SHANKAR DIGHE COLLEGE OF SCIENCE

(Affiliated to University of Mumbai)

CERTIFICATE BY THE INSTITUTE

Second Year B.Sc. Computer Science – 2026/27

Certificate

I hereby certify that Mr. Krushna Gupta, Student of V. K. Krishna Menon College, studying in Third Year B.Sc. Computer Science program has completed a Community Engagement Project titled “Air and Noise Pollution Mapping” in the area of Environmental Science and Community Engagement for the academic year 2026–2027. To the best of my knowledge the work of the student is original and the information included in the project is correct.

Internal Guide

Head of Department

Principal

STUDENT'S DECLARATION

I, Mr. Krushna Gupta, Student of V. K. Krishna Menon College, studying in Second Year B.Sc. Computer Science program, hereby declare that I have completed the Community Engagement Project titled "Air and Noise Pollution Mapping" during the academic year 2026–2027.

The report is original and the information/data included in the report is true emerging from the primary and secondary data gathered and analyzed as part of this Community Engagement Project.

Due credit is extended on the work of Literature/Secondary Survey by endorsing it in the Bibliography as per prescribed format.

Signature of the Student: _____

Name: Krushna Gupta

Date: _____

Place: Dombivli

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who guided and supported me throughout the completion of this Community Engagement Project.

First and foremost, I thank the University of Mumbai for incorporating the Community Engagement Project component under NEP 2020, providing students like me an opportunity to connect academic learning with real-world community issues.

I am deeply grateful to my Faculty Mentor, Mr. Rakesh for her continuous guidance, constructive feedback, and unwavering encouragement throughout the project. Her insights helped shape the direction and quality of this work significantly.

I extend my heartfelt thanks to the Principal and the Head of the Department of V. K. Krishna Menon College for providing the necessary resources, infrastructure, and permissions to carry out this project.

I am truly thankful to the residents, shopkeepers, teachers, and local volunteers of Dombivli, who welcomed our team, answered our survey questions, shared their experiences, and participated enthusiastically in our awareness activities. Their cooperation was invaluable.

I also thank my classmates and friends who accompanied me during the field measurements and made the experience both productive and memorable.

Finally, I express my profound gratitude to my family for their constant motivation and moral support throughout this journey.

Krushna Gupta

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ABSTRACT

This Community Engagement Project (CEP) report documents a series of field-based activities undertaken by Krushna Gupta (Roll No. 11, TY BSc CS) in partial fulfilment of the curriculum requirements under NEP 2020. The project focuses on two interconnected environmental themes — Air Pollution and Noise Pollution — and on the use of simple digital tools to map them across selected locations in Dombivli, Thane district, Maharashtra.

Air and noise pollution are among the most persistent yet least visible urban environmental problems. Rapid urbanisation, dense road traffic, construction activity, and industrial clusters have raised exposure levels in cities of the Mumbai Metropolitan Region, but most residents have little access to locality-level information. This project was designed to engage with this gap by combining low-cost field measurement, community interaction, and awareness-driven volunteering, and by presenting the results in the form of easily understandable pollution maps.

Three major community interactions were conducted: (1) a baseline air and noise monitoring survey at seven sampling locations, (2) a community perception survey and interview session with residents and shopkeepers, and (3) an awareness campaign and pollution-map presentation at a residential society and a school. The methodology included observational research, low-cost sensor measurements, GPS-tagged data collection, semi-structured interviews, secondary data review, and data visualisation using Python-based mapping tools.

Key findings indicate that measured daytime noise levels exceeded the prescribed CPCB limits at six of the seven sites, and short-duration particulate matter readings were above the National Ambient Air Quality Standards benchmark at four to five sites, with the highest values near the industrial periphery and the busiest traffic junctions. The community survey of 60 respondents showed that while 82% noticed dust or smoke regularly, only 27% were familiar with the Air Quality Index. The awareness campaign reached about 120 residents and students.

The project concludes that community-based engagement, when backed by academic learning and simple technology, has the potential to make invisible pollution visible, foster environmental sensitivity, and support meaningful local action.

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Chapter 1: Introduction

1.1 Purpose of the Visit

This Community Engagement Project was undertaken with the primary objective of bridging the gap between academic learning in B.Sc. Computer Science and real-world environmental concerns affecting the city of Dombivli and the wider Mumbai Metropolitan Region. The selected area of focus, Air and Noise Pollution Mapping, aligns directly with the Sustainable Development Goals (SDG 3: Good Health and Well-being, SDG 11: Sustainable Cities and Communities, and SDG 13: Climate Action) and with NEP 2020's vision of socially engaged students.

The specific objectives of this project were:

- To measure and document air quality (PM2.5 and PM10) and ambient noise levels at selected locations in Dombivli representing different land-use zones.
- To compare the measured values with the standards prescribed in India and the guidelines of the World Health Organization.
- To present the collected data as easy-to-read pollution maps using simple digital and GIS tools.
- To understand how residents perceive air and noise pollution and how aware they are of existing standards and remedies.
- To conduct an awareness campaign that shares the findings with the community and encourages responsible behaviour.
- To develop research, data-handling, observation, and communication skills through field-based work.

1.2 Background Information

The Mumbai Metropolitan Region is one of the most densely populated urban clusters in India. Dombivli, part of the Kalyan-Dombivli Municipal Corporation area in Thane district, has grown rapidly as a residential and commercial suburb, with a busy railway station, crowded market roads, continuing construction activity, and an industrial estate (MIDC) in close proximity to residential neighbourhoods.

Air pollution, especially fine particulate matter (PM2.5 and PM10), can penetrate deep into the respiratory system and is linked to respiratory and cardiovascular illness. Noise pollution, though often treated as a mere nuisance, is associated with sleep disturbance, stress, poor concentration, and cardiovascular effects. Both problems are strongly linked to road traffic, construction, industrial activity, waste burning, and the use of loudspeakers and horns. Despite this, ordinary residents rarely have access to locality-level information on how polluted their surroundings are.

Given the background in Computer Science, an additional dimension of this project was to explore how low-cost sensors, spreadsheets, and open-source mapping tools can be used to collect, analyse, and visualise environmental data, and how digital communication can help share the results with the community — reflecting the interdisciplinary spirit of NEP 2020.

1.3 Scope of the Report

This report documents three community interaction sessions conducted between January 2026 and March 2026. It covers:

- Theoretical background on air pollution, noise pollution, and environmental mapping.
- A detailed account of each community interaction, including location, participants, activities, and outcomes.
- Measurement data collected during field visits and the maps and charts derived from it.
- Reflections on learning, challenges faced, and the impact of the activities.
- Concrete recommendations for further community engagement in this area.

The report does not claim regulatory-grade accuracy. Measurements were made with low-cost instruments over short durations and are indicative only; the report does not attempt a full-scale scientific air quality or noise survey. The scope is defined by the community engagement and learning dimension of the project.

Chapter 2: Literature Review

2.1 Air Pollution: Health Effects and Standards

Particulate matter is classified by size: PM₁₀ refers to particles of 10 micrometres or less in diameter, and PM_{2.5} to particles of 2.5 micrometres or less. The smaller particles are of greater health concern because they can reach the lungs and enter the bloodstream. The Lancet Commission on pollution and health (Landrigan et al., 2018) estimated that pollution was responsible for around nine million premature deaths in 2015, about one in six deaths worldwide, with air pollution the largest contributor and low- and middle-income countries, including India, the worst affected.

In India, the Central Pollution Control Board (CPCB) notified the National Ambient Air Quality Standards (NAAQS) in 2009. For PM_{2.5} the standard is 40 µg/m³ as an annual average and 60 µg/m³ as a 24-hour average; for PM₁₀ it is 60 µg/m³ annually and 100 µg/m³ over 24 hours. The World Health Organization's global air quality guidelines (2021) are considerably stricter, recommending 5 µg/m³ (annual) and 15 µg/m³ (24-hour) for PM_{2.5}, and 15 µg/m³ (annual) and 45 µg/m³ (24-hour) for PM₁₀. To communicate air quality to the public, CPCB launched the National Air Quality Index (AQI) in 2014, which converts pollutant concentrations into six categories from 'Good' to 'Severe'. The National Clean Air Programme (2019) aims to reduce particulate pollution in a large number of Indian cities, including several in the Mumbai Metropolitan Region.

2.2 Noise Pollution: Impacts and Regulation

Basner et al. (2014) reviewed the evidence on the effects of noise on health and concluded that exposure to environmental noise, particularly at night, is associated with sleep disturbance, annoyance, cognitive impairment in children, and increased risk of cardiovascular disease. The WHO Environmental Noise Guidelines for the European Region (2018) recommend keeping average road-traffic noise below 53 dB Lden and night-time noise below 45 dB Lnight.

In India, noise is regulated under the Noise Pollution (Regulation and Control) Rules, 2000, framed under the Environment (Protection) Act, 1986. The Rules prescribe ambient noise limits for four area categories. The daytime (6 a.m. to 10 p.m.) and night-time (10 p.m. to 6 a.m.) limits are 75/70 dB(A) for industrial areas, 65/55 dB(A) for commercial areas, 55/45 dB(A) for residential areas, and 50/40 dB(A) for silence zones. Silence zones are areas within 100 metres of hospitals, educational institutions, courts, and religious places notified as such. The Rules also restrict the use of loudspeakers and public address systems at night and regulate the use of horns.

2.3 Pollution Mapping and Low-Cost Sensing

Mapping converts scattered measurements into a form that non-experts can quickly understand. The European Union's Environmental Noise Directive (2002/49/EC) requires member states to produce strategic noise maps for large agglomerations and major roads, and air quality maps are widely used for planning and public information. Kumar et al. (2015) describe the rise of low-cost sensing

technologies that make it possible to monitor air pollution at a much finer spatial resolution than sparse regulatory monitoring stations, while cautioning that such sensors require careful calibration and that their data should be interpreted as indicative.

Citizen science — the involvement of volunteers in scientific data collection — has been recognised as an effective tool for expanding scientific knowledge and public scientific literacy (Bonney et al., 2009). Smartphone-based sound level applications, open-source mapping libraries, and freely available GIS software have made it feasible for students and community groups to carry out simple mapping exercises of the kind attempted in this project.

2.4 Awareness Campaigns and Behaviour Change

Kollmuss and Agyeman (2002) identified a gap between environmental knowledge and pro-environmental behaviour, noting that effective campaigns must address attitudinal, motivational, and structural barriers, not just provide information. Successful campaigns typically combine multiple formats — visual media, interactive workshops, field experiences, and community storytelling. Maps and local data are particularly persuasive because they make the problem personal: residents respond more readily to information about their own street than to national averages.

2.5 The Dombivli Context

Dombivli has a mix of dense residential areas, busy market roads, and an industrial estate located close to housing. The MIDC area has repeatedly featured in news reports because of industrial pollution complaints and accidents, which has kept the question of local environmental quality in public discussion. However, publicly available, locality-level air and noise data for individual streets and neighbourhoods remain limited, which is the gap this project attempts to address in a small way.

Chapter 3: Methodology

3.1 Research Design and Approach

This project adopted a mixed-method, participatory action research approach. The study combined quantitative methods (instrument-based measurement of PM_{2.5}, PM₁₀, and noise levels; counting of survey responses; documenting audience sizes) with qualitative methods (field observations, semi-structured interviews, reflective journaling). This approach was chosen because community engagement requires both measurable outcomes and a depth of understanding of how people experience the problem.

The overall design follows a three-phase structure:

1. **Pre-field preparation:** Literature review, objective setting, selection of sampling sites, preparation and testing of instruments and data sheets, and planning of interactions.
2. **Field execution:** Three community interaction sessions spread across January to March 2026.
3. **Post-field analysis:** Data cleaning, comparison with standards, preparation of maps and charts, compilation of field notes, photographs and interview summaries, and writing of the report.

3.2 Data Collection Methods

The following methods were employed for data collection:

- **Air quality measurement:** PM_{2.5} and PM₁₀ concentrations were recorded using a portable low-cost laser-scattering particulate sensor. At each site, readings were logged for 15 minutes and averaged.
- **Noise measurement:** Equivalent continuous sound level (Leq) in dB(A) was recorded for 15 minutes at each site using a smartphone sound-level application, checked against a reference sound level meter before the survey to estimate the offset.
- **GPS tagging:** The latitude and longitude of each site was noted using a smartphone so that readings could be placed on a map.
- **Observational research:** Systematic observation of traffic density, honking, idling vehicles, construction activity, waste burning, and other visible sources, recorded in a field notebook.
- **Structured survey and semi-structured interviews:** A short questionnaire was administered to residents and shopkeepers, and brief informal interviews were held with a traffic constable, a shop owner, a local doctor, a school teacher, and a member of a housing society committee.
- **Secondary data review:** CPCB standards, the Noise Rules, WHO guidelines, and published research papers were reviewed to contextualise field observations.
- **Photography:** Photographs were taken at the field sites to document conditions and activities.

3.3 Study Area and Sampling Locations

The study area comprised the Dombivli (East and West) locality of Kalyan-Dombivli. Seven sampling locations were selected to represent different land-use zones as defined in the Noise Rules, so that the effect of zone type on pollution levels could be compared. Each location was visited in three time slots — morning (8–10 a.m.), afternoon (1–3 p.m.) and evening (6–8 p.m.) — to capture variation through the day. The sites are listed in Table 3.1.

Code	Sampling Location	Zone Type	Daytime Noise Limit dB(A)
S1	Dombivli Railway Station Approach Road (West)	Commercial / high traffic	65
S2	Phadke Road Market Area (East)	Commercial	65
S3	Kopar Road Junction	Commercial / mixed	65
S4	College Campus, Telcos Wadi	Silence zone (educational institution)	50
S5	Residential Lane near Kopar Road	Residential	55
S6	Hospital–School Zone (West)	Silence zone	50
S7	MIDC Phase II Periphery	Industrial	75

Table 3.1: Sampling Locations and Zone Classification

[Sample entry — replace site names with the locations you actually visited.]

3.4 Data Processing and Mapping

All readings were entered into a spreadsheet along with site code, GPS coordinates, time slot, and observations. The data were cleaned, averaged for each site, and compared with the NAAQS 24-hour benchmarks, WHO 2021 guideline values, and the noise limits for the relevant zone. A short Python script using the pandas and folium libraries was written to plot the sites on an interactive web map, with markers colour-coded by category (within limit, moderately above limit, and well above limit). Bar charts were generated using matplotlib. The sample script is given in Appendix II.

3.5 Ethical Considerations and Limitations

Informed consent was obtained verbally from all survey and interview participants. No personally identifiable information about community members has been disclosed in this report. Photographs were taken with the awareness of those present, and no photographs of individuals are included without their consent. All measurements were made in public places without obstructing traffic or pedestrians, and the team followed safety practices near roads and the industrial periphery.

The main limitations of the study are: (i) low-cost instruments have lower accuracy than regulatory monitors; (ii) each reading lasted only 15 minutes, so values are not equivalent to 24-hour averages and comparison with NAAQS benchmarks is indicative only; (iii) measurements were taken on a

limited number of days and are influenced by weather, traffic, and local events; and (iv) the sample of seven sites and 60 survey respondents is small and not statistically representative of the whole city.

Chapter 4: Community Interactions, Observations and Analysis

4.1 Overview of Community Interactions

Three major community interactions were conducted as part of this project. A summary is presented in Table 4.1 below:

No.	Activity	Date & Location	Participants	Key Outcome
1	Baseline Air and Noise Monitoring Survey	22 Jan 2026, seven sites across Dombivli	8 students	21 sets of readings (7 sites × 3 time slots) and a baseline dataset
2	Community Perception Survey and Interviews	12 Feb 2026, Phadke Road area and nearby housing societies	12 students, 60 residents and shopkeepers, 5 interviewees	60 questionnaire responses and 5 interview summaries
3	Map Presentation and Eco-Awareness Campaign	12 Mar 2026, residential society and school	~120 residents and students reached	Pollution map poster, pamphlets, quiz, and interactive session

Table 4.1: Summary of Community Interactions

[Sample entry — replace dates, places, and numbers with your actual details.]

4.2 Interaction 1: Baseline Air and Noise Monitoring Survey (22 January 2026)

Description:

The first community interaction was a field survey in which a team of eight students, divided into two groups, visited the seven pre-selected sampling locations in Dombivli. At each location, the team recorded PM2.5, PM10, and noise level for 15 minutes in each of the three time slots, noted the GPS coordinates, and recorded visible pollution sources. Members of the public who asked about the activity were briefed on its purpose.

Observations:

Table 4.2 presents the mean values recorded at each site, and Figures 4.1 and 4.2 compare them with the relevant benchmarks.

Site	Location	PM2.5 (µg/m³)	PM10 (µg/m³)	Noise Leq dB(A)	Noise Limit for Zone dB(A)
S1	Station Approach Road (W)	88	165	76	65
S2	Phadke Road Market (E)	74	142	73	65
S3	Kopar Road Junction	69	131	70	65
S4	College Campus, Telcos Wadi	46	92	58	50
S5	Residential Lane near Kopar Road	41	84	54	55
S6	Hospital–School Zone (W)	52	101	61	50
S7	MIDC Phase II Periphery	97	189	79	75

Table 4.2: Air and Noise Measurements at Sampling Locations (mean of three 15-minute readings)

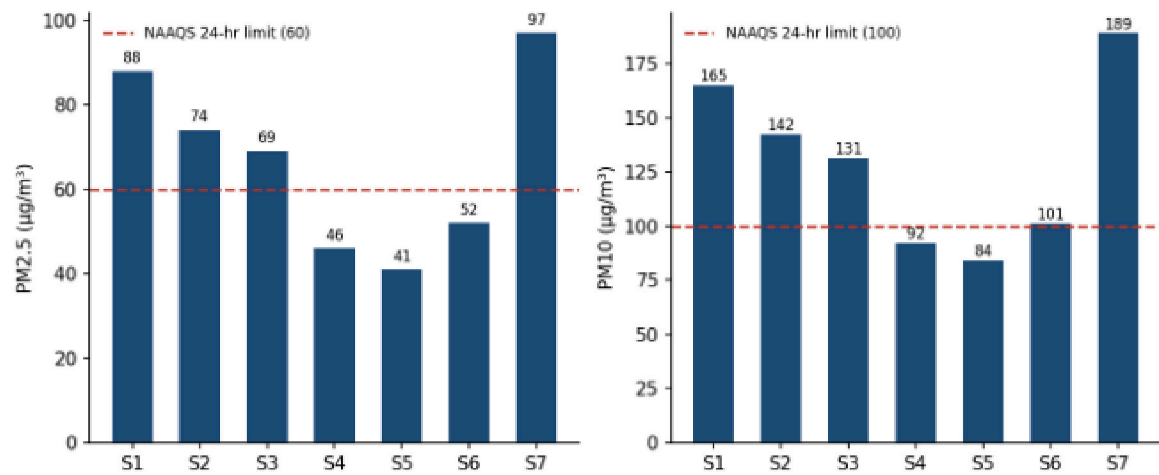


Figure 4.1: Mean PM2.5 and PM10 Levels against NAAQS Benchmarks

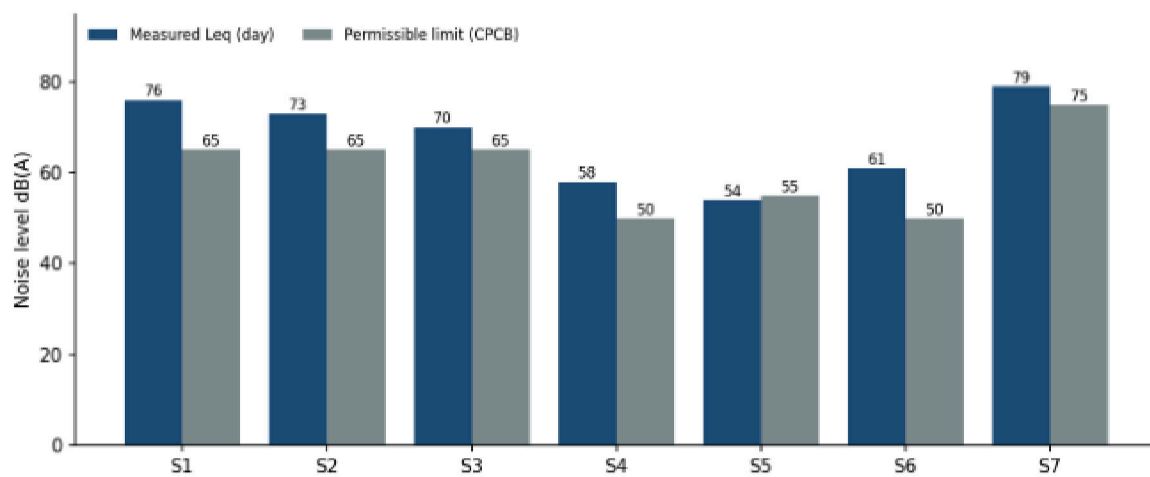


Figure 4.2: Measured Daytime Noise Levels against Permissible Limits

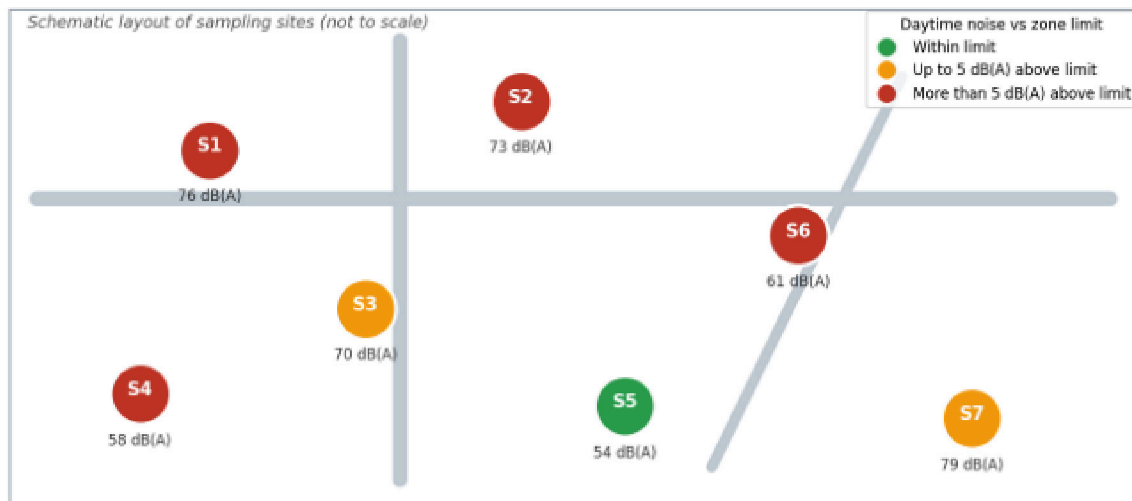


Figure 4.3: Schematic Map of Sampling Sites Colour-Coded by Noise Level against Zone Limit

Analysis:

The measurements clearly showed that pollution levels varied with land use and traffic intensity. The highest particulate levels and noise levels were recorded at the MIDC periphery (S7) and at the station approach road (S1), both of which were dominated by heavy vehicular movement, idling vehicles, and, in the case of S7, visible dust and smoke from the industrial periphery. The lowest values were recorded in the residential lane (S5), which was set back from the main road.

Measured daytime noise exceeded the prescribed limit at six of the seven sites, including both silence-zone locations (S4 and S6), where the campus and the hospital–school zone recorded levels 8 to 11 dB(A) above the limit because of traffic on the adjoining roads and frequent honking. Only the residential lane (S5) remained within its limit. Noise was consistently higher during the morning and evening peak hours than in the afternoon; at S1, for example, the morning and evening readings were about 6 dB(A) higher than the afternoon reading.

For particulate matter, four sites (S1, S2, S3 and S7) exceeded the NAAQS 24-hour benchmark for PM_{2.5} (60 µg/m³), and five sites (those four plus S6) exceeded the benchmark for PM₁₀ (100 µg/m³). All seven sites were above the stricter WHO 24-hour PM_{2.5} guideline of 15 µg/m³. It must be stressed that these were short-duration readings and are indicative only. Other visible sources noted during the survey included resuspended road dust, construction activity near S3, and burning of roadside waste near the S7 periphery.

This interaction deepened my understanding of the direct link between urban traffic, land use, and the local environment, and highlighted how little locality-level information is available to residents — reinforcing the importance of the perception survey and awareness campaign that followed.

4.3 Interaction 2: Community Perception Survey and Interviews (12 February 2026)

Description:

The second community interaction was a perception survey conducted in the Phadke Road area and in nearby housing societies. Twelve students administered a short questionnaire to 60 residents and shopkeepers, and five semi-structured interviews were held with a traffic constable, a shop owner, a local doctor, a school teacher, and a member of a housing society committee. Participation was voluntary and consent was taken verbally.

Observations:

The main results of the questionnaire are summarised in Table 4.3.

Statement / Question	No. of Respondents	Percentage
Consider road traffic (engines and horns) the main source of noise	41	68%
Notice dust or smoke in the air at least once a week	49	82%
Report disturbed sleep or difficulty concentrating due to noise	35	58%
Had heard of the Air Quality Index (AQI)	16	27%
Know that silence zones and noise limits exist	7	12%
Willing to take part in community monitoring or awareness activities	43	72%

Table 4.3: Results of Community Perception Survey (n = 60)

In the interviews, the traffic constable and the shop owner both pointed to continuous honking and illegal parking as major contributors to noise at the market and junction. The local doctor said that patients with breathing problems often mentioned dust and smoke as triggers, without being able to identify their source. The school teacher described difficulty in conducting lessons near the main road, and the housing society member reported that complaints about construction dust and late-night loudspeakers rarely reach the concerned authorities because residents do not know where or how to complain.

Analysis:

The survey showed that residents are clearly aware of pollution as a lived problem — most notice dust or smoke weekly and more than half reported effects on sleep or concentration — but that awareness of the tools and rules that could help them is low. Only about one in four respondents had heard of the AQI and very few knew of silence zones or noise limits. This knowledge gap matched the observation in the literature that information alone is not enough, but that its absence is a real barrier to action.

At the same time, nearly three-quarters of respondents were willing to participate in community monitoring or awareness activities, which suggests strong potential for a citizen-science approach. Interacting directly with residents also gave me an appreciation of the socio-economic dimension of

the problem: shopkeepers and street vendors, who spend the longest hours at the most polluted locations, are also the least able to avoid exposure.

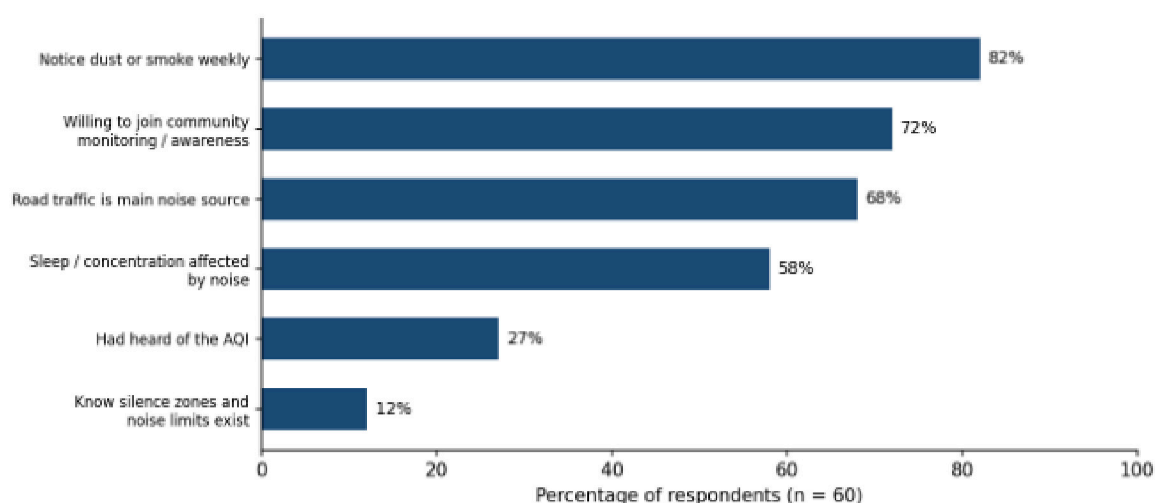


Figure 4.4: Key Results of the Community Perception Survey

4.4 Interaction 3: Map Presentation and Eco-Awareness Campaign (12 March 2026)

Description:

The third interaction was an awareness campaign conducted at a residential housing society and an adjacent school in Dombivli. The campaign was designed and implemented by our student team, leveraging digital skills to create pollution maps, informational posters, short videos, and interactive quizzes. The event ran for approximately 4 hours and reached an estimated audience of 120 residents and students.

Activities conducted:

- Distribution of bilingual (Marathi–English) pamphlets explaining the health effects of air and noise pollution, the meaning of the AQI, permissible noise limits, and simple steps residents can take.
- Display of a poster exhibition showing the pollution maps and charts produced from the survey, with a QR code linking to the interactive web map.
- Screening of a short self-edited video (3 minutes) documenting the measurement survey, shared on the society’s WhatsApp group.
- A live demonstration of how to read a particulate sensor and a smartphone sound-level application.
- Interactive quiz session for school students (Classes 6–8) on air and noise pollution, with prizes for winners, followed by a “No Unnecessary Honking” pledge.

Air Quality Index (CPCB) categories

Good 0-50	Satisfactory 51-100	Moderate 101-200	Poor 201-300	Very Poor 301-400	Severe 401-500
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Permissible ambient noise limits, dB(A) (Noise Rules, 2000)

Area	Day (6 am-10 pm)	Night (10 pm-6 am)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence zone	50	40

Figure 4.5: AQI Categories and Noise Limits Used as Reference in the Campaign Material

Analysis:

The awareness campaign was the most publicly visible component of the project and required the most preparation. Designing effective communication material for a diverse, multilingual urban audience was a challenge — one that directly connected our Computer Science skills (data handling, visualisation, graphic design, video editing, digital distribution) to community service.

The response from the audience was encouraging. Most residents were surprised to see measured values for their own neighbourhood and asked whether their own street was on the map. Several asked how they could check the AQI regularly and where to complain about noise violations. The school quiz session was particularly effective — children were energetic and competitive, and their enthusiasm demonstrated that environmental education is highly receptive when delivered in interactive formats.

A key learning from this interaction was that maps make invisible problems visible. Colour-coded maps communicated the findings far faster than tables of numbers, and the short video shared digitally was reported to have been forwarded to over 100 people in the following week — well beyond our physical reach. This demonstrates how digital literacy, when applied to community causes, can amplify impact significantly.

Chapter 5: Conclusion and Recommendations

5.1 Summary of Key Findings

This Community Engagement Project, conducted over three major interactions between January and March 2026, yielded the following key findings:

1. **Pollution varies sharply with land use:** Traffic junctions, market roads, and the industrial periphery recorded the highest particulate and noise levels, while the residential lane recorded the lowest.
2. **Noise limits are widely exceeded:** Daytime noise exceeded prescribed limits at six of seven sites, including both silence zones, with clear morning and evening peaks associated with traffic and honking.
3. **Particulate levels are a concern:** Short-duration PM2.5 and PM10 readings were above the NAAQS 24-hour benchmarks at several sites, and all sites were above the WHO guideline values (indicative readings only).
4. **Awareness is low but willingness is high:** Most residents noticed dust and smoke regularly, yet few knew about the AQI or noise limits, while a large majority were willing to participate in community monitoring.
5. **Maps and technology amplify awareness:** Colour-coded maps, short videos, mobile sharing, and interactive quizzes communicated the findings effectively and extended the reach of the engagement beyond physical boundaries.
6. **Personal learning:** The project developed research, data-handling, communication, teamwork, and leadership skills that classroom learning alone cannot provide, and reinforced a sense of civic responsibility and environmental sensitivity.

5.2 Contribution to Subject Understanding

As a student of Computer Science, this project offered a unique opportunity to explore the intersection of technology and environmental sustainability. It demonstrated real-world applications of data collection, cleaning and analysis, geospatial visualisation, and digital communication. The experience of collecting sensor readings, processing them with Python, and presenting them as interactive maps is directly relevant to fields such as data science, environmental informatics, smart-city applications, and the Internet of Things — all increasingly important domains in the technology industry.

The project also broadened my understanding of interdisciplinary problem-solving. Air and noise pollution are not solved by engineers or scientists alone — they require policy intervention, enforcement, urban planning, community mobilisation, and technology solutions working in concert.

5.3 Recommendations

Based on the observations, community interactions, and analysis conducted in this project, the following recommendations are offered:

1. **Establish a Community Monitoring Network:** A group of students, residents, and volunteers should be trained to take regular readings at fixed locations, so that a continuous local dataset can be built over time.
2. **Deploy Low-Cost Sensors with Open Dashboards:** Calibrated low-cost sensors at key junctions, schools, and hospitals could feed a public web dashboard showing live AQI and noise levels, a low-cost, high-impact application of technology.
3. **Enforce Silence Zones and Anti-Honking Measures:** Visible signage, periodic enforcement of noise limits near schools, hospitals, and colleges, and “no honking” campaigns at major junctions should be strengthened in coordination with the traffic police and the municipal corporation.
4. **Control Dust and Waste Burning:** Construction sites should be encouraged to follow dust-control measures such as barricading and water sprinkling, and open burning of waste should be discouraged and reported.
5. **Simplify Public Grievance Channels:** Residents should be informed about where and how to report noise and air pollution violations, and pollution-related complaint mechanisms should be made simple and widely publicised.
6. **Annual Awareness Events:** The awareness campaign should be institutionalised as an annual event, co-organised by the college and local groups, timed to coincide with World Environment Day (June 5) or Clean Air Day.
7. **Policy Advocacy and Further Research:** Students and faculty should collectively prepare a report with the maps and findings for submission to the concerned local authorities, and a follow-up study with longer measurement periods and more sites should be undertaken to measure long-term changes in pollution and awareness.

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APPENDIX I

GUIDE INTERACTION DIARY FORM

I, the undersigned Mr. Krushna Gupta, Roll No. 52, currently enrolled in the Third Year of B.Sc. Computer Science Full-time Program at V. K. Krishna Menon College, am undertaking my Community Engagement Project work under the guidance of Mr. Rakesh.

and I hereby confirm that I have met my Internal guide on the following dates mentioned below for Project Guidance:

Sr. No.	Date	Signature of the Internal Guide
1		
2		
3		
4		
5		

Signature of the Candidate

Signature of Internal Guide

IMPORTANT: It is expected that students will be meeting their guide at least five times for the CEP work interaction. The candidate should retain the above stated 'Project Guide Interaction Certificate Form' and submit the same with required signatures of the guide while submitting the Project to the Institute.

THE PROJECT REPORT WILL NOT BE ACCEPTED WITHOUT THE DULY FILLED PROJECT GUIDE INTERACTION CERTIFICATE.

APPENDIX II

SAMPLE MAPPING SCRIPT

The following Python script reads a CSV file of readings (columns: site, lat, lon, pm25, pm10, leq_db, limit_db) and produces an interactive map with colour-coded markers. It requires the pandas and folium libraries.

```
import pandas as pd
import folium

df = pd.read_csv("readings.csv")
m = folium.Map(location=[df.lat.mean(), df.lon.mean()], zoom_start=14)

def noise_colour(leq, limit):
    if leq <= limit:
        return "green"
    return "orange" if leq - limit <= 5 else "red"

for _, r in df.iterrows():
    popup = (f"{r.site}<br>PM2.5: {r.pm25} ug/m3<br>"
             f"PM10: {r.pm10} ug/m3<br>Noise: {r.leq_db} dB(A)")
    folium.CircleMarker(
        location=[r.lat, r.lon], radius=10,
        color=noise_colour(r.leq_db, r.limit_db),
        fill=True, popup=popup).add_to(m)

m.save("pollution_map.html")
```

ANNEXURE V

Student Feedback on Community Engagement Project

(To be filled by Student after CEP completion)

Student Name: Krushna Gupta

Seat No. / Roll No.: 52

Email: _____

Department: B.Sc. Computer Science

Name of the Mentor: Mr. Rakesh

Title/Heading of CEP: Air and Noise Pollution Mapping

Brief description of CEP: Field-based community engagement involving air and noise pollution measurement at sites in Dombivli, a community perception survey, and a mapping-based awareness campaign.

Dates of CEP: July 2026 – October 2026

My CEP experience was related to my major area of study / academic program:

☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree

Indicate the degree to which you agree or disagree with the following statements:

This experience has:	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Increased my sensitivity towards societal problems	✓				
Allowed me to apply classroom theory to practice	✓				
Helped me develop decision-making and problem-solving skills	✓				
Helped me develop written and oral communication skills	✓				
Provided a chance to use leadership skills	✓				
Expanded my sensitivity to ethical implications	✓				
Made me more confident in new situations	✓				
Helped me improve interpersonal skills	✓				
Helped me learn to handle responsibility	✓				
Helped me discover new aspects of myself	✓				
Allowed me to realize socio-economic issues in society	✓				

The faculty mentor extended guidance and mentoring throughout the CEP process:

☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree

Were you able to accomplish the initial goals and skills set in CEP plan?

☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree

Considering your overall experience, how would you rate this CEP?

☐ Poor ☐ Satisfactory ☐ Good ☐ Excellent

Suggestions to improve CEP experience:

Aspect of CEP Experience	Satisfactory	Needs Improvement
Clear orientation and briefing at the start	✓	
Relevance of project to academic learning	✓	
Availability and support from internal guide	✓	
Adequate field exposure and hands-on engagement	✓	
Clarity in roles and responsibilities	✓	
Time provided for the completion of project		X

Signature of Student: _____

Name: Krushna Gupta

Date: _____